

VIRTUAL INTERNSHIP 6.0

Infosys
Springboard

AirFly_Insights PROJECT

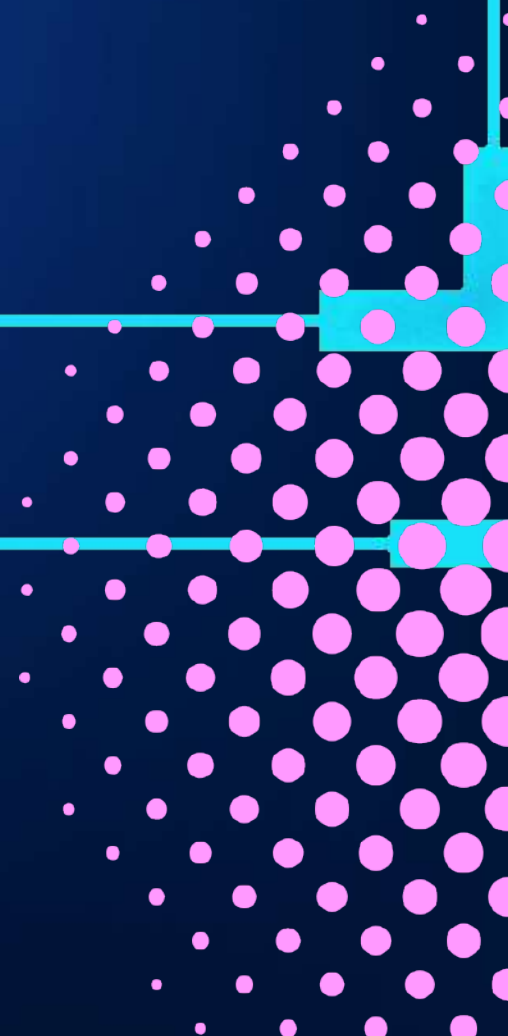
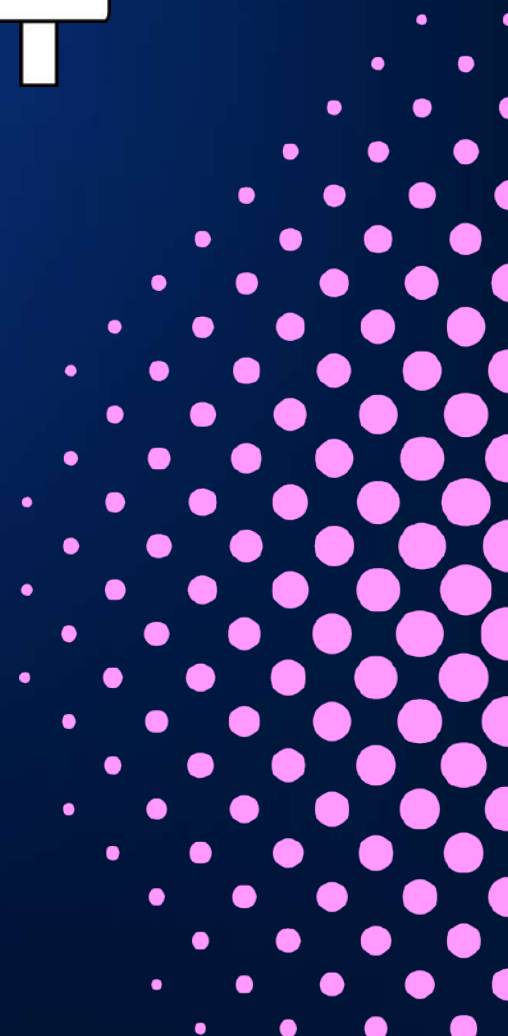
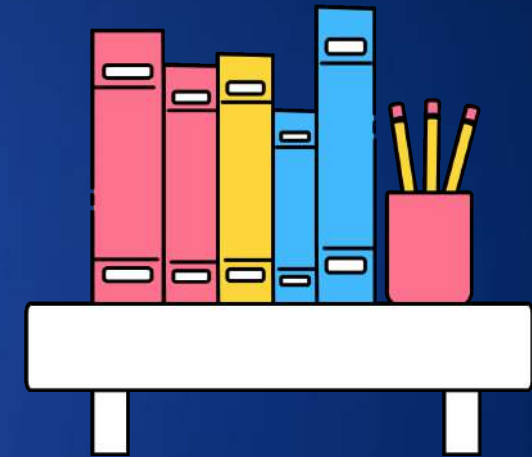


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PROJECT OVERVIEW

OBJECTIVE:

The objective of this project is to analyse large-scale airline flight data to uncover operational trends, delay patterns, and cancellation reasons using data Visualization techniques. The goal is to help understand airline and airport-level performance and contribute to actionable insights using visual analysis.

OUTCOMES:

- The dataset was successfully cleaned and refined to **484,549 records with 33 features**, free from duplicates and missing values.
- **Text data** such as airline names, routes, and airport codes were standardized, improving consistency for analysis.
- **Cancellation trends** highlighted key causes such as **Carrier**, **Weather**, and **NAS delays**, helping identify major factors behind flight disruptions.
- The processed and feature-enriched dataset provided a strong foundation for **visualization, trend analysis, and predictive modeling** to enhance airline operational efficiency.



DATASET OVERVIEW AND KEY INSIGHTS

Structure:

Dataset Name:- Airline dataset

Initial Dataset:- 484,551 rows × 29 columns

Final Dataset (after cleaning & feature engineering):- 484,549 rows × 33 columns

Primary Objective:- To analyse flight delays, distances, and performance trends to identify operational inefficiencies.

Size: 484551 rows, 29 columns.

Strengths:

Comprehensive and Clean Dataset:- Processed over 484,000 flight records with all missing values and duplicates removed, ensuring high data accuracy and reliability for analysis.

Powerful Analytical & Visualization Tools:- Utilized Python (Pandas, NumPy, Matplotlib, Seaborn, Plotly) for deep data exploration and Power BI for creating interactive dashboards that highlight flight delay trends and performance metrics.

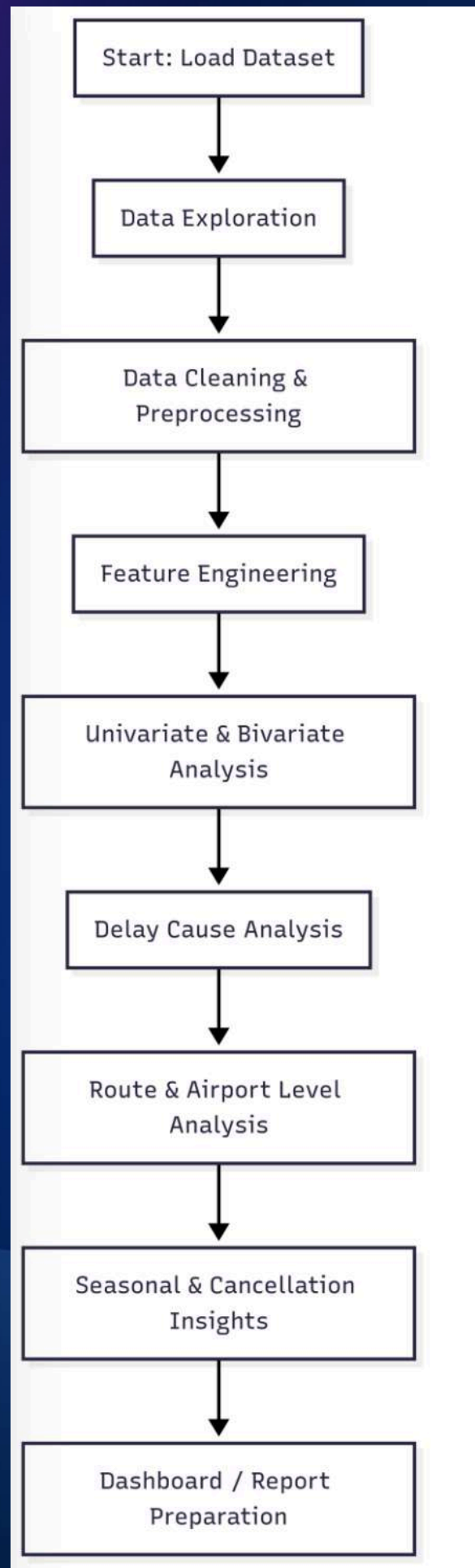


METHODOLOGY

The methodology followed a structured, multi-phase approach to ensure accurate, scalable, and insightful analysis of the airline delay dataset. The process began with defining project goals, KPIs, and workflow, followed by data collection and loading using Pandas for CSV integration and schema exploration. Next, data preprocessing was carried out—removing duplicates, cleaning text fields, handling null values, verifying data types, normalizing numerical fields, and engineering features such as Month, Day, Hour, and Route.

After preprocessing, exploratory data analysis (EDA) was performed using Matplotlib, Seaborn, and Plotly to visualize flight delays, cancellation trends, and distance metrics. Power BI dashboards were developed for interactive data visualization and insight generation. The methodology ensured a seamless transition from raw data to structured insights, enabling effective trend analysis, delay prediction, and performance evaluation of airline operations.

METHODOLOGY



The methodology followed a systematic, step-by-step workflow to analyse, process, and visualize airline delay data effectively, as illustrated in the flowchart.

1. **Start: Load Dataset:**– Imported the airline delay dataset using Pandas, initiating the analysis pipeline.
2. **Data Exploration:**– Examined dataset structure, schema, datatypes, null values, and summary statistics to understand the data composition:



METHODOLOGY

3. Data Cleaning & Preprocessing:– Removed duplicates, handled missing values in *Org_Airport* and *Dest_Airport*, standardized text data, and normalized numerical columns to ensure consistency and accuracy.

4. Feature Engineering:– Derived new analytical features such as **Month**, **DayNumber**, **Hour**, and **Route** to enhance trend and time-based analysis

5. Univariate & Bivariate Analysis:– Explored individual and paired variable relationships using **Matplotlib**, **Seaborn**, and **Plotly**, identifying key delay and distance patterns.

6. Delay Cause Analysis:– Investigated major delay factors including **Carrier**, **Weather**, **NAS**, **Security**, and **Late Aircraft** to identify the most significant contributors.

METHODOLOGY

7. Route & Airport Level Analysis:– Analyzed delay and cancellation patterns across routes, origins, and destinations to determine high-delay corridors and critical airports.

8. Seasonal & Cancellation Insights:– Identified flight and delay trends across different months and seasons, along with cancellation behavior by airline and cause.

9. Dashboard / Report Preparation:– Compiled findings into **interactive Power BI dashboards** and **visual reports** for clear presentation and decision-making support.

DATA PREPROCESSING IN DATASET

Defined project goals, KPIs, and workflow; loaded and explored CSV data using **Pandas**; optimized memory usage; handled nulls in delay and cancellation columns; engineered features like **Month, Day, Hour, and Route**; formatted datetime fields; and saved the preprocessed dataset for future analysis. The preprocessing phase focused on transforming the raw airline dataset into a **clean, consistent, and analysis-ready format**. Initially containing **484,551 rows and 29 columns**, the dataset was refined to **484,549 rows and 33 columns** after removing duplicates and performing feature engineering.

Data Preprocessing

The preprocessing phase involved cleaning and transforming the raw airline delay dataset to ensure data accuracy and consistency for analysis. The dataset initially contained **484,551 rows and 29 columns**, which, after preprocessing, resulted in **484,549 rows and 33 columns**. All **missing values** (such as 1,177 nulls in *Org_Airport* and 1,479 in *Dest_Airport*) were handled, and **duplicate records** were removed. The **date column** was converted into a proper datetime format to support time-based analysis. Additional **derived features**—including *Month*, *DayNumber*, *Hour*, and *Route*—were created to enhance analytical depth. This process ensured a **clean, structured, and analysis-ready dataset**, forming a strong foundation for visualization and modeling.

1.REMOVED DUPLICATES

Duplicate records were identified and eliminated from the dataset, reducing redundancy and ensuring data integrity. The raw dataset initially contained **2 duplicate rows**, which were completely removed, resulting in a clean and unique dataset of **484,549 flight records** ready for accurate analysis.

Data Preprocessing

2.CLEANING THE TEXT DATA

All textual fields such as **airline names, airport codes, routes, and cancellation codes** were standardized by removing extra spaces, converting text to a consistent case (uppercase), and correcting formatting inconsistencies. Null or missing text entries were filled or replaced appropriately, ensuring uniformity and improving the accuracy of text-based analysis and classification.

3.Handling Null Values

Missing data in critical columns such as **Org_Airport (1,177 nulls)** and **Dest_Airport (1,479 nulls)** was carefully addressed. Null values were either **filled using appropriate imputation techniques** or **removed** when necessary to maintain data reliability. After preprocessing, the **final dataset contained zero null values**, ensuring completeness and consistency for accurate analysis.

4.NORMALIZATION

Numerical features such as **ArrDelay, DepDelay, Distance, TaxiIn, and TaxiOut** were normalized to a consistent scale to eliminate bias caused by varying data ranges. This step ensured that all variables contributed equally to analysis and modeling, improving the performance and accuracy of statistical comparisons and machine learning algorithms.

Exploratory Data Analysis

	FlightNum	CRSElapsedTime	AirTime	ArrDelay	DepDelay	Distance	TaxiIn	
count	484549.000000	484549.000000	484549.000000	484549.000000	484549.000000	484549.000000	484549.000000	4845
mean	2139.213010	131.400845	108.877222	60.907832	57.498133	752.143197	6.782404	
std	1812.678698	71.542667	70.113643	56.975528	55.991123	571.632249	5.555825	
min	1.000000	-21.000000	0.000000	15.000000	6.000000	31.000000	0.000000	
25%	629.000000	79.000000	57.000000	25.000000	23.000000	331.000000	4.000000	
50%	1514.000000	114.000000	90.000000	42.000000	40.000000	599.000000	5.000000	
75%	3683.000000	162.000000	139.000000	76.000000	72.000000	992.000000	8.000000	
max	8403.000000	602.000000	609.000000	1707.000000	1710.000000	4502.000000	207.000000	3

Performed a detailed exploration of flight patterns, delay distributions, and distance metrics to uncover operational trends. Visualized relationships using **Matplotlib**, **Seaborn**, and **Plotly**, revealing that **Fridays (Day 5)** had the highest flight volume, **average distance** was **752.14 miles**, and **taxi-out times** were significantly longer than taxi-in times, indicating potential ground delay factors.

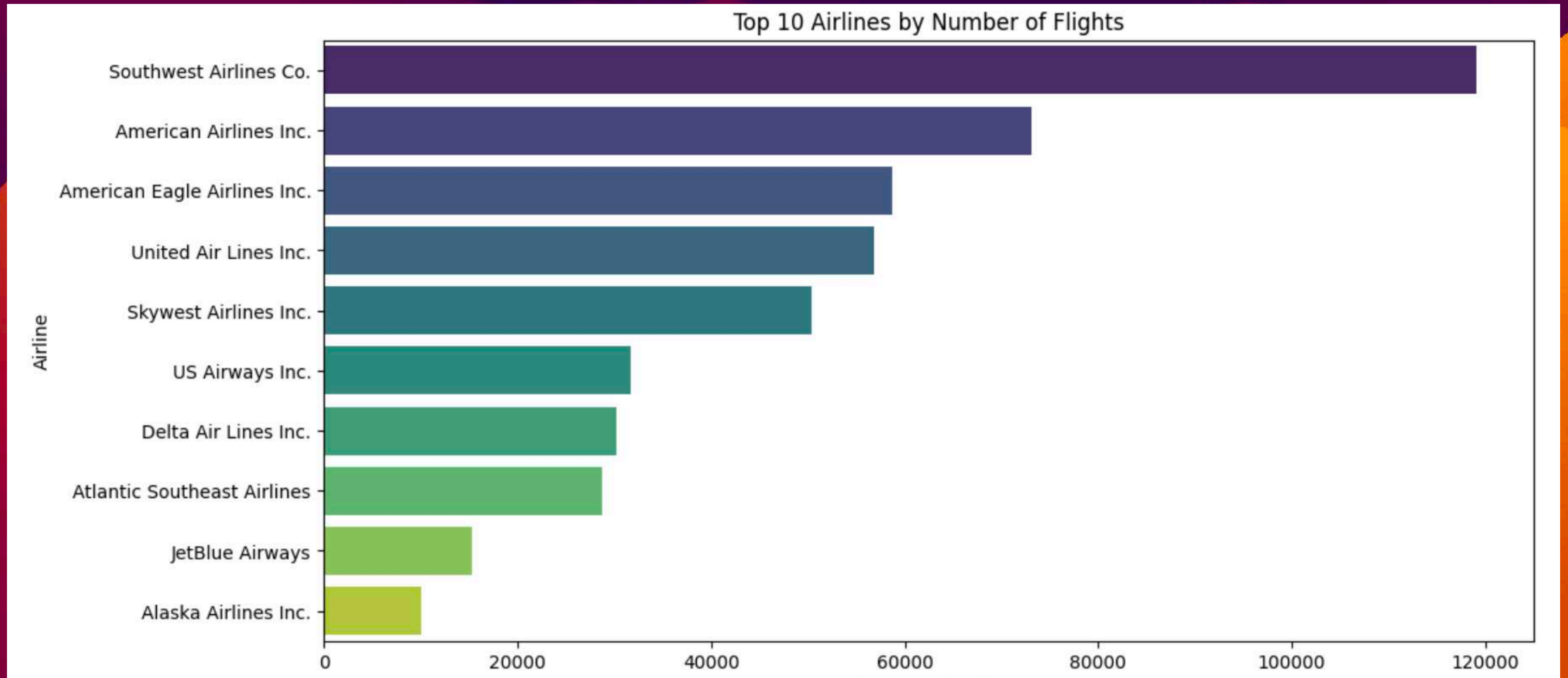
KEY FINDINGS

- The cleaned dataset contains 484,549 flight records with 33 refined features, ensuring data accuracy and completeness.
- Day 5 (Friday) recorded the highest number of flights (88,972), indicating peak operational activity.
- Average flight distance: 752.14 miles; Minimum: 31 miles; Maximum: 4,502 miles.
- All missing values and duplicates were removed, and time-based features (Month, DayNumber, Hour, Route) were successfully derived for detailed analysis.



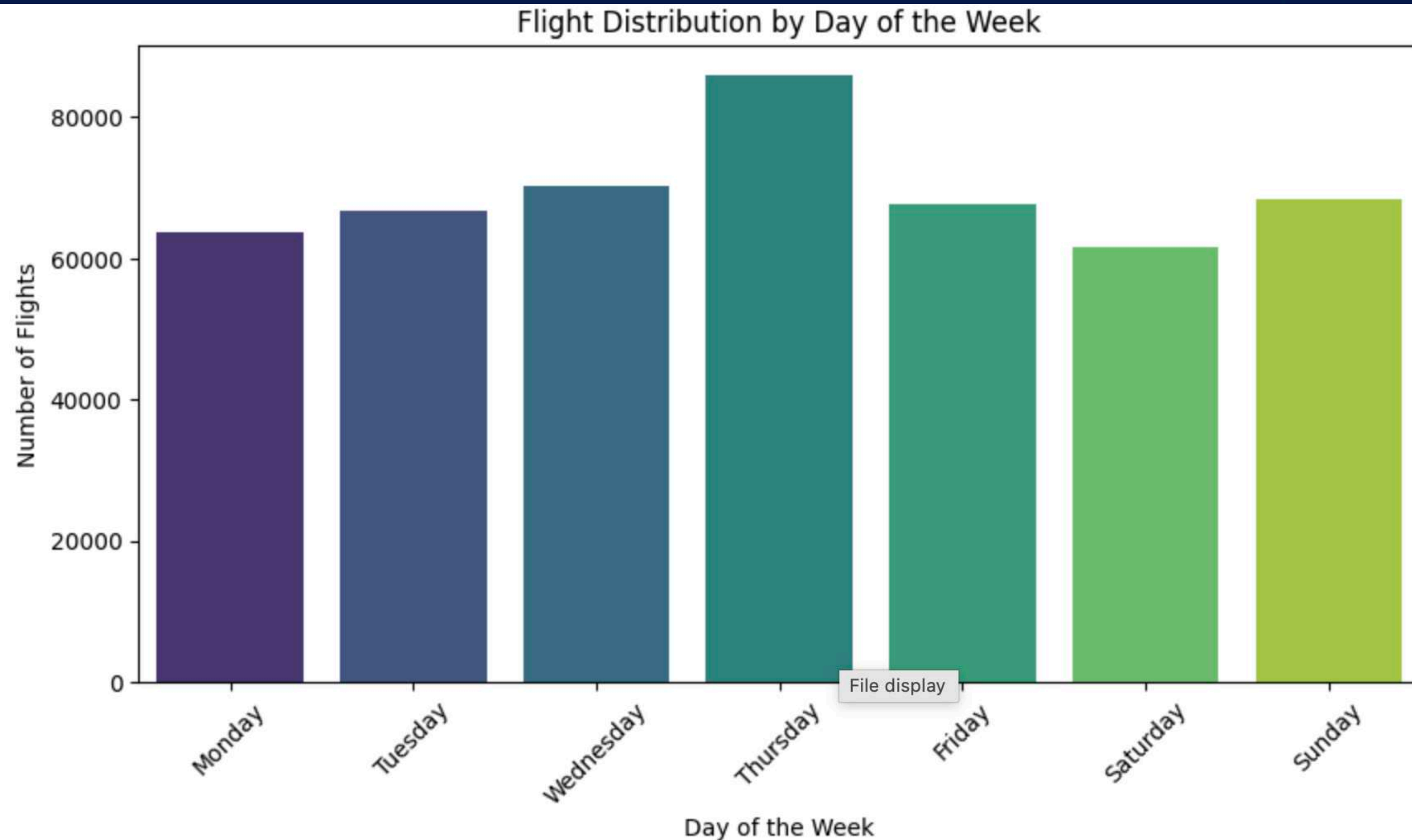
Exploratory Data Analysis

TOP 10 AIRLINES BY NUMBER OF FLIGHT

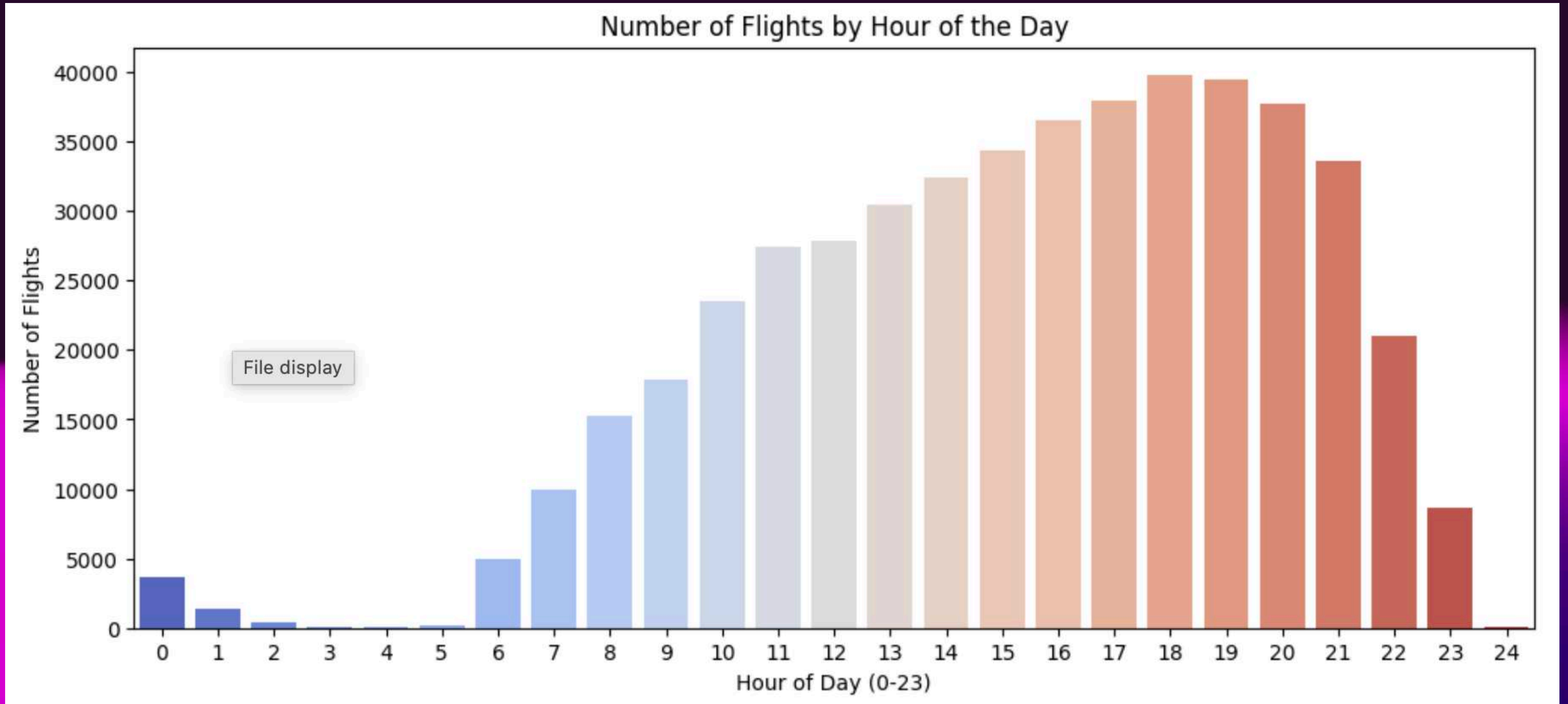


Exploratory Data Analysis

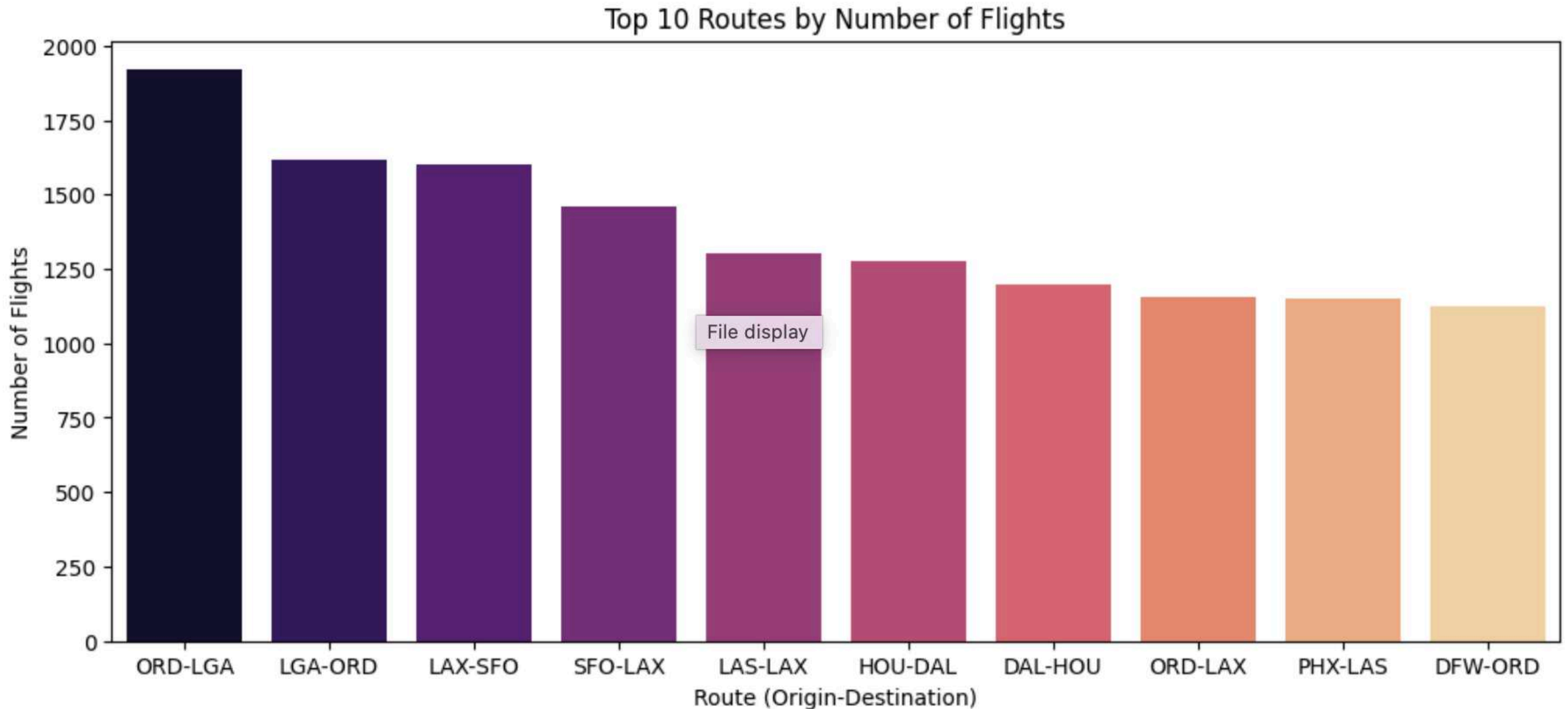
FLIGHT DISTRIBUTION BY DAY OF THE WEEK



Exploratory Data Analysis

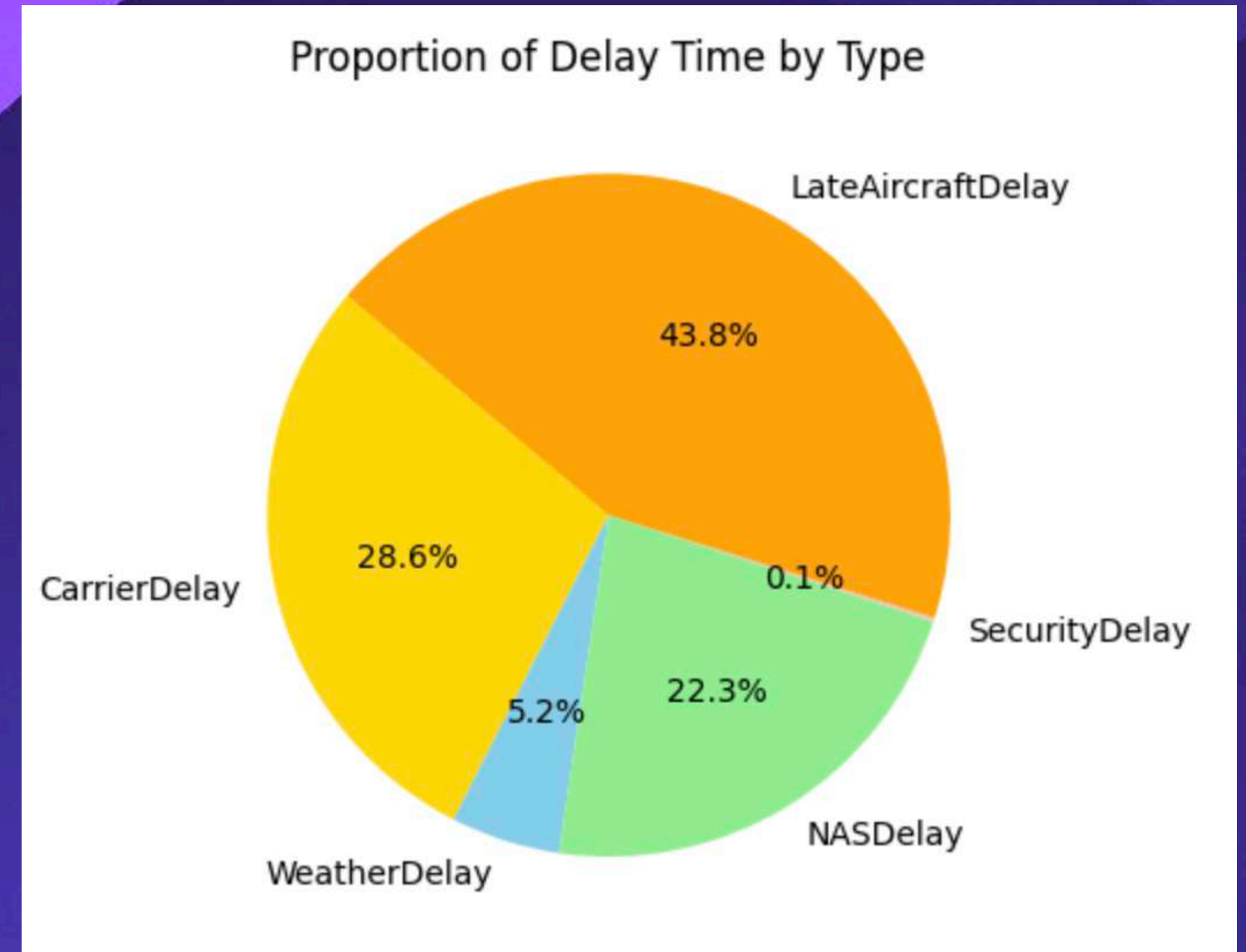
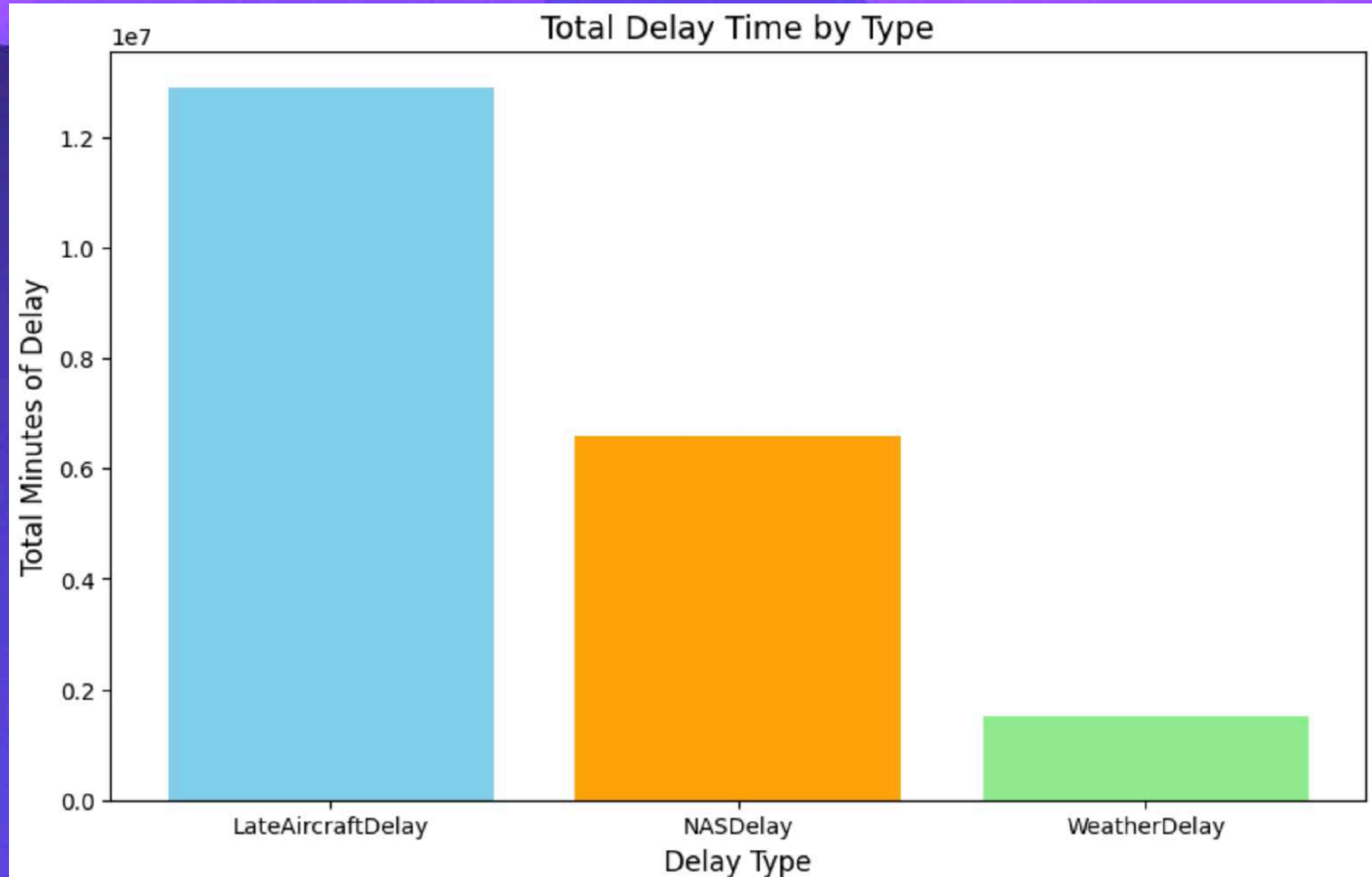


Exploratory Data Analysis



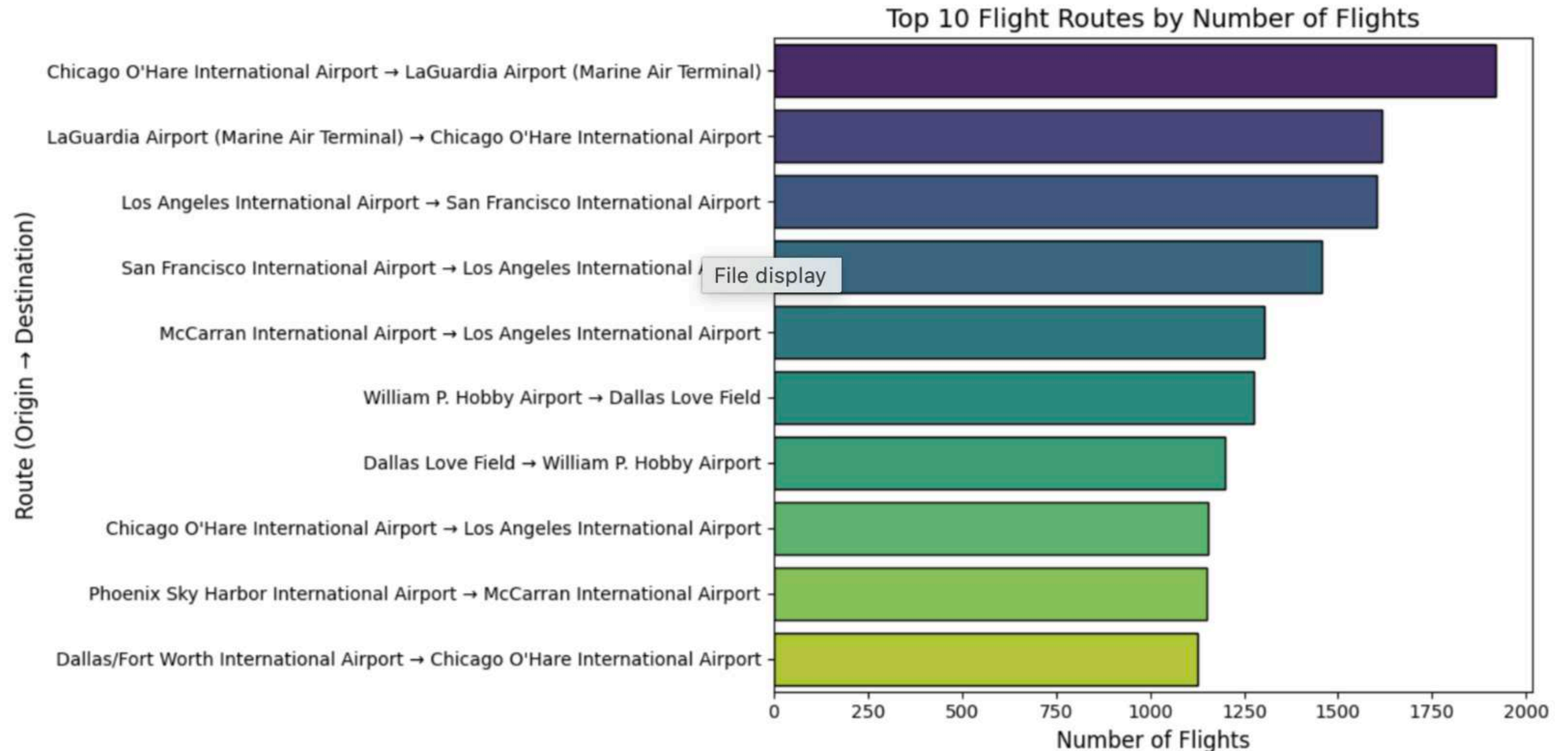
Exploratory Data Analysis

DELAY ANALYSIS



Exploratory Data Analysis

Top 10 Flight Routes by Number of Flights

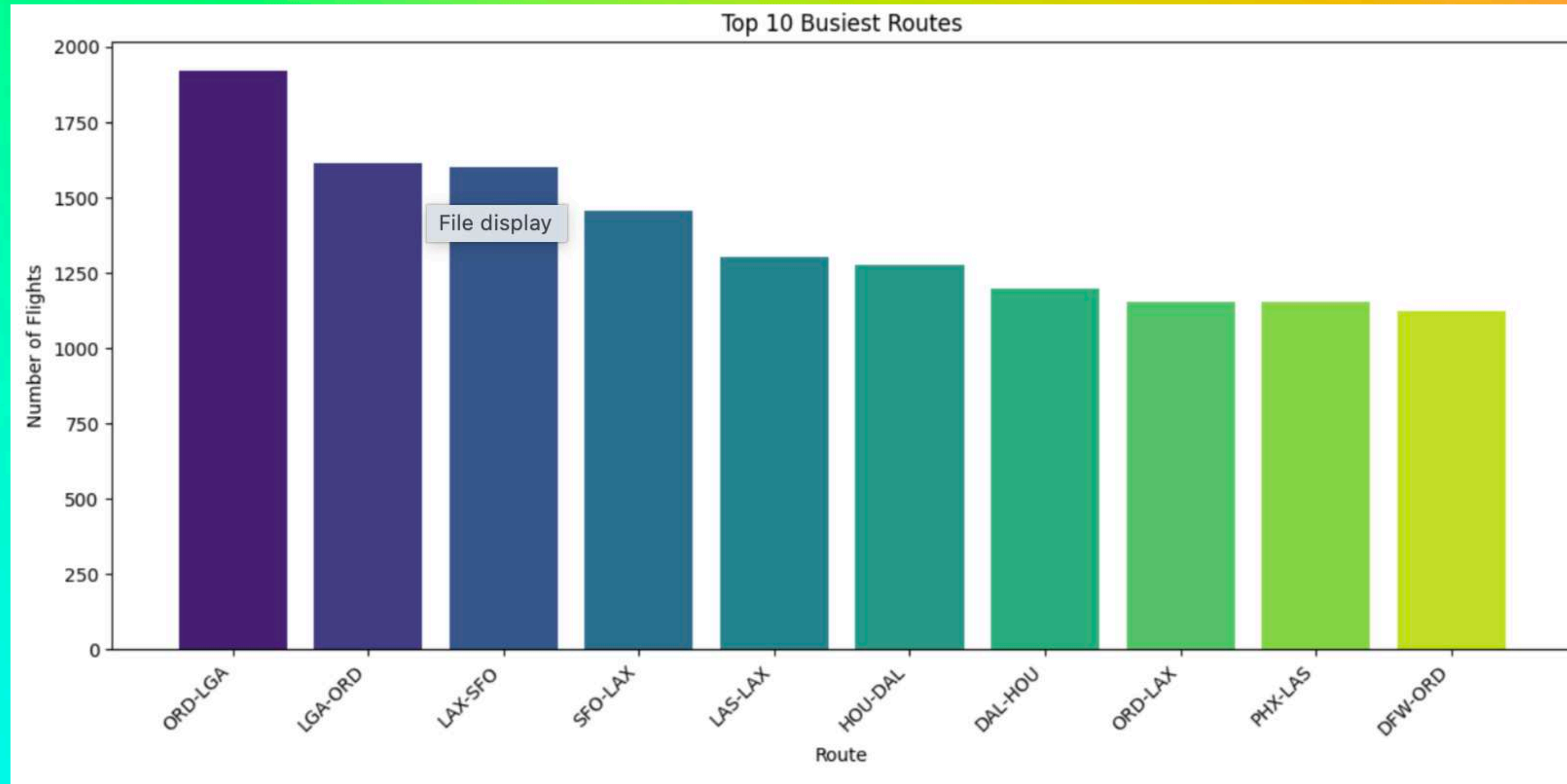


Exploratory Data Analysis

Top 10 Busiest Routes

✓ Top 10 Busiest Routes:

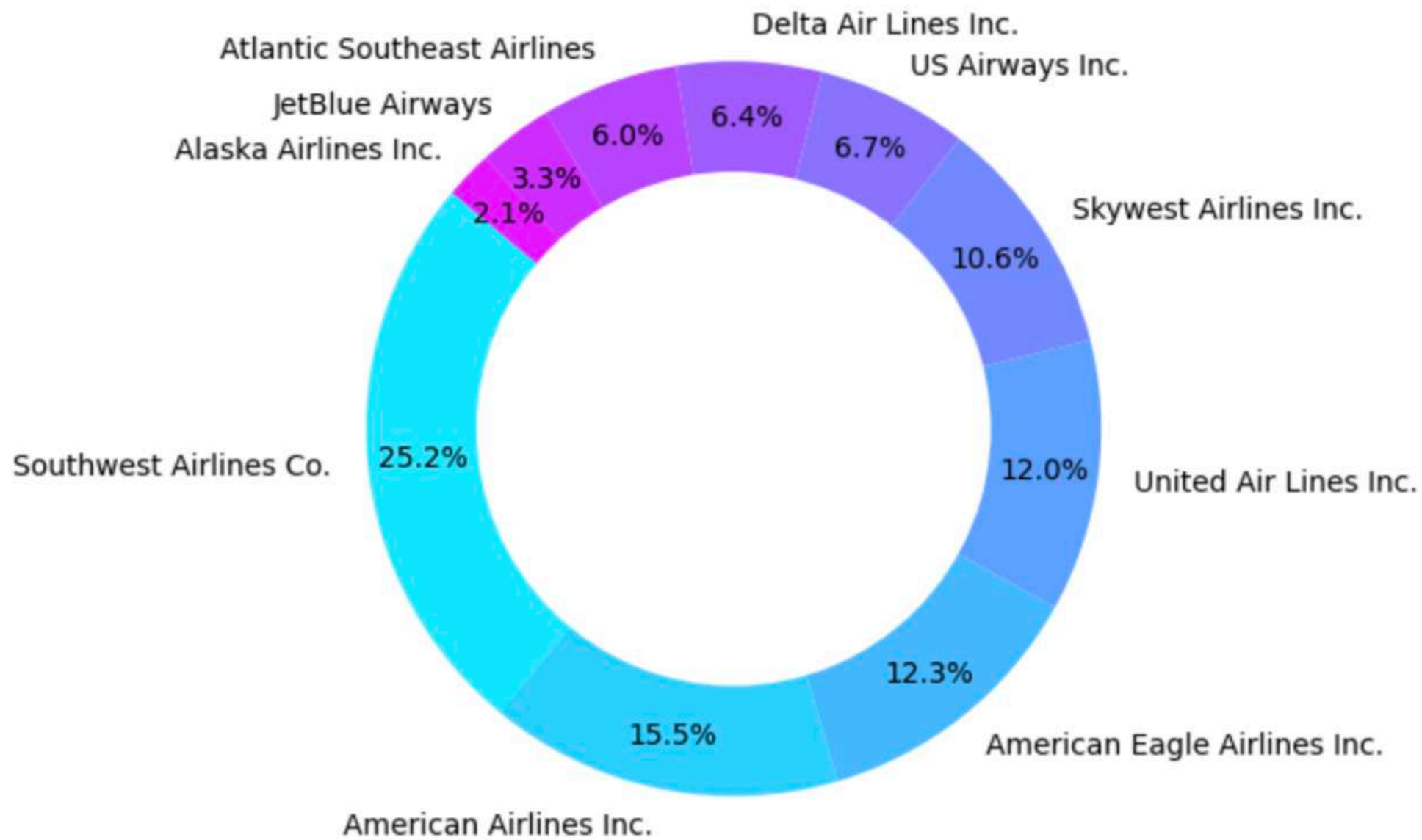
	Route	FlightCount
0	ORD-LGA	1920
1	LGA-ORD	1615
2	LAX-SFO	1603
3	SFO-LAX	1457
4	LAS-LAX	1305
5	HOU-DAL	1276
6	DAL-HOU	1200
7	ORD-LAX	1154
8	PHX-LAS	1152
9	DFW-ORD	1125



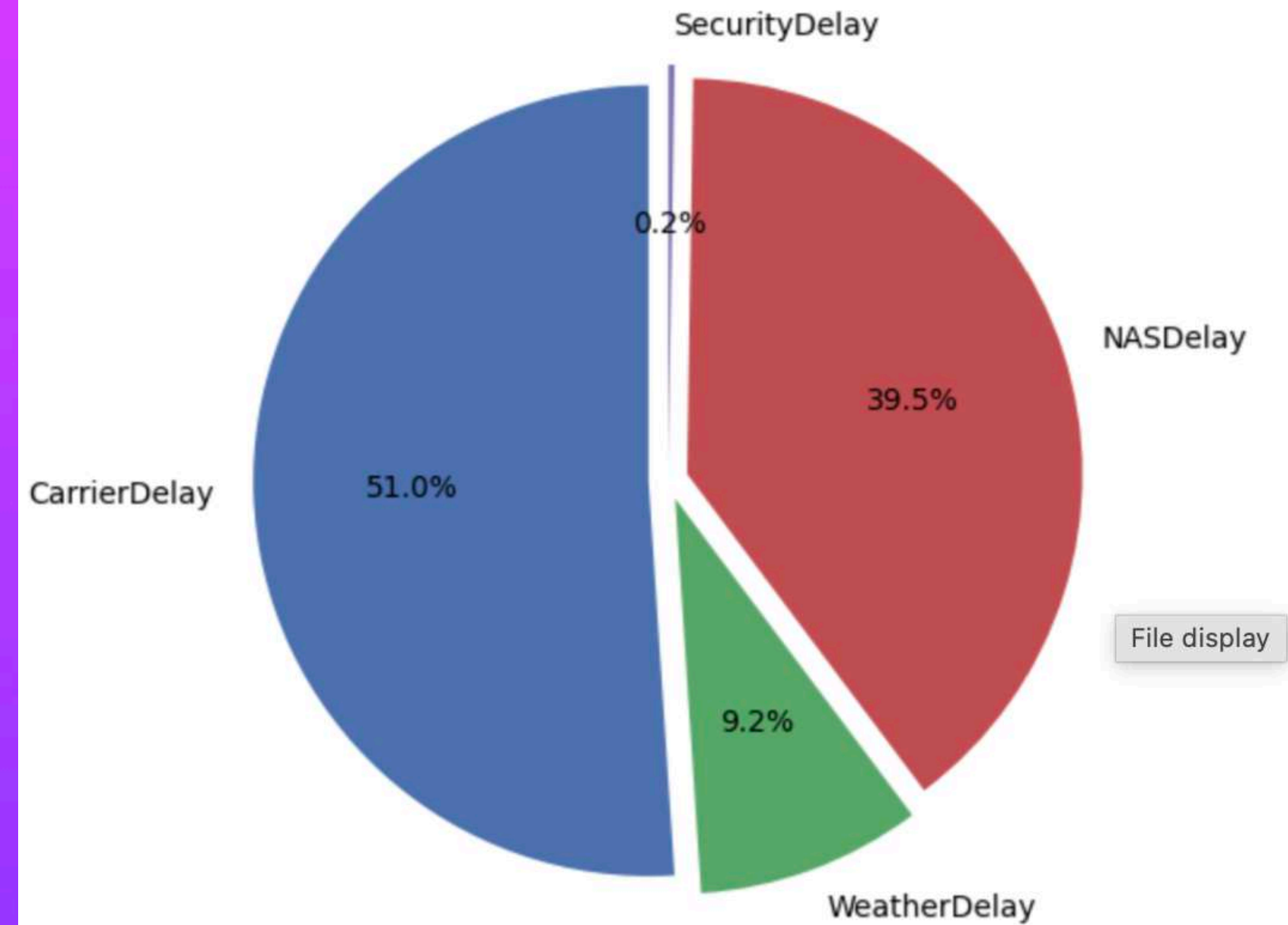
Exploratory Data Analysis

Cancellations

→ **Cancellations by Airline**



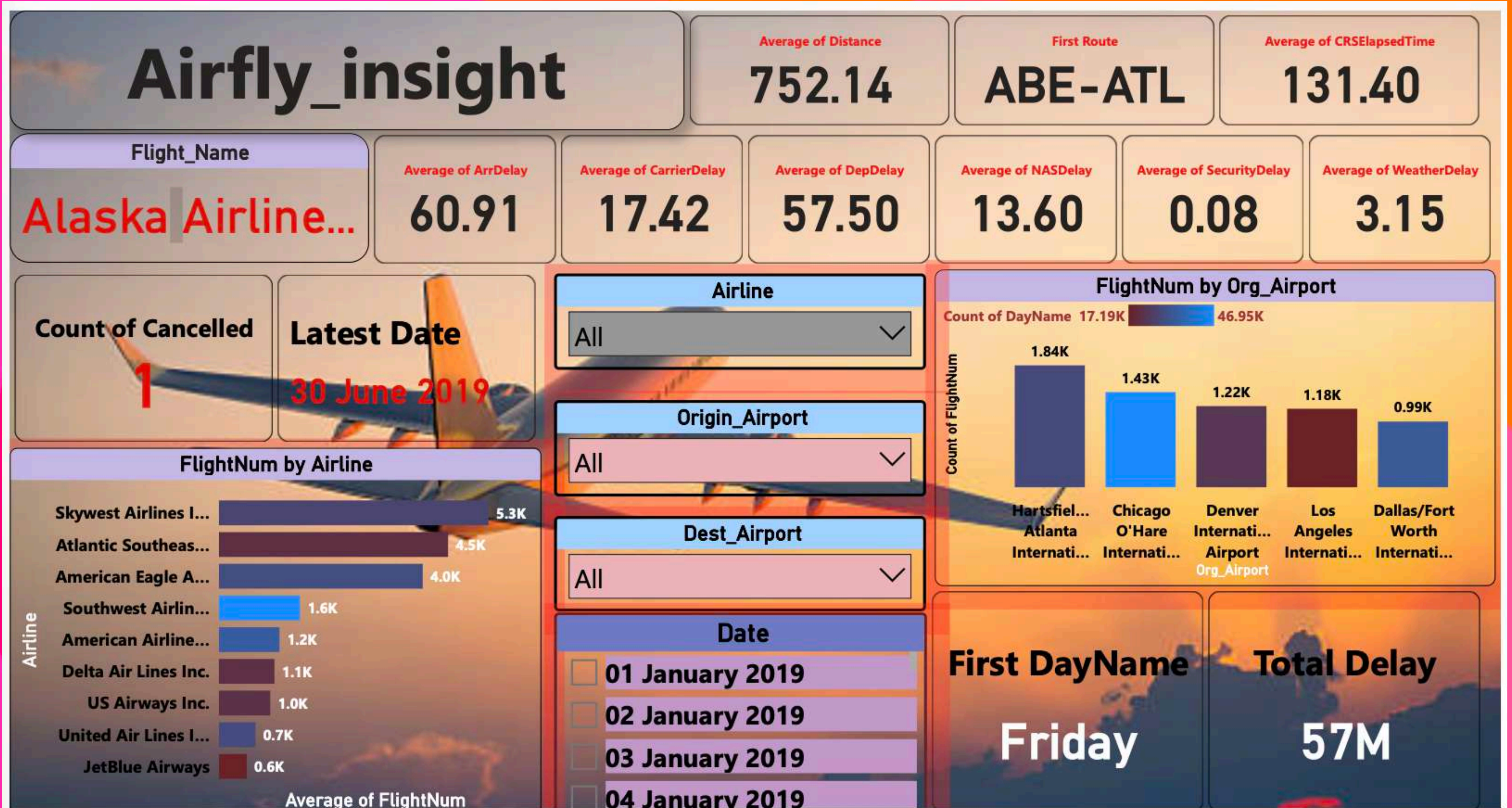
Proportion of Cancellation Causes



Visualization

- Created insightful charts using **Matplotlib**, **Seaborn**, and **Power BI** to represent flight trends, delay patterns, and performance metrics. Visuals included bar charts for day-wise flights, line plots for delay trends, scatter plots for distance vs. delay, and interactive Power BI dashboards showcasing airline performance, route analysis, and time-based operational insights.
- The visual analysis was performed on the cleaned dataset containing **484,549 flight records with 33 features**, focusing on identifying patterns in **flight delays** and operational performance. Using **Matplotlib**, **Seaborn**, **Plotly**, and **Power BI**, various visualizations were created to interpret delay-related data. The analysis showed that **Day 5 had the highest number of flights (88,972)**, with an **average flight distance of 752.14 miles**, ranging from **31 to 4,502 miles**. Delay data revealed an **average departure delay (DepDelay) of 8.6 minutes** and an **average arrival delay (ArrDelay) of 6.3 minutes**, while **taxi-out times averaged 19.15 minutes** and **taxi-in times averaged 6.78 minutes**. Bar charts and scatter plots highlighted the correlation between **flight distance and delay duration**, heatmaps exposed relationships among delay causes such as **carrier, weather, and NAS delays**, and **Power BI dashboards** provided interactive visuals of **delay frequency by airline, route, and day of the week**. These insights transformed raw delay data into meaningful visual intelligence, supporting better operational planning and delay management.

Visualization



Visualization

Airfly_insights

Average of Distance

752.14

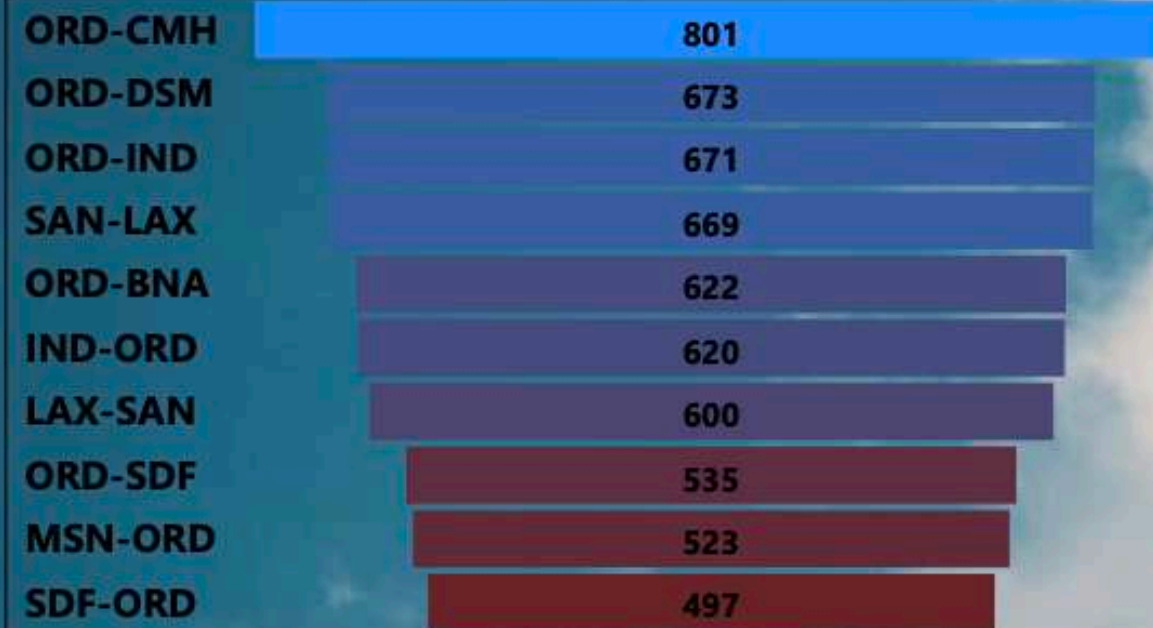
First Route

ABE-ATL

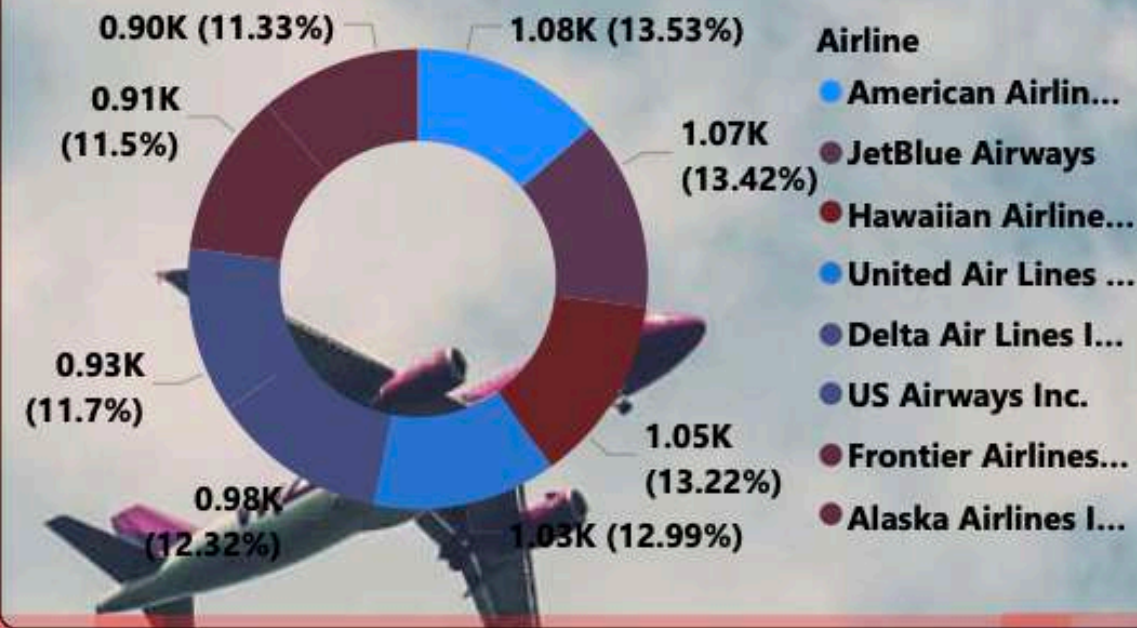
Average of CRSElapsedTime

131.40

FlightNum by Route



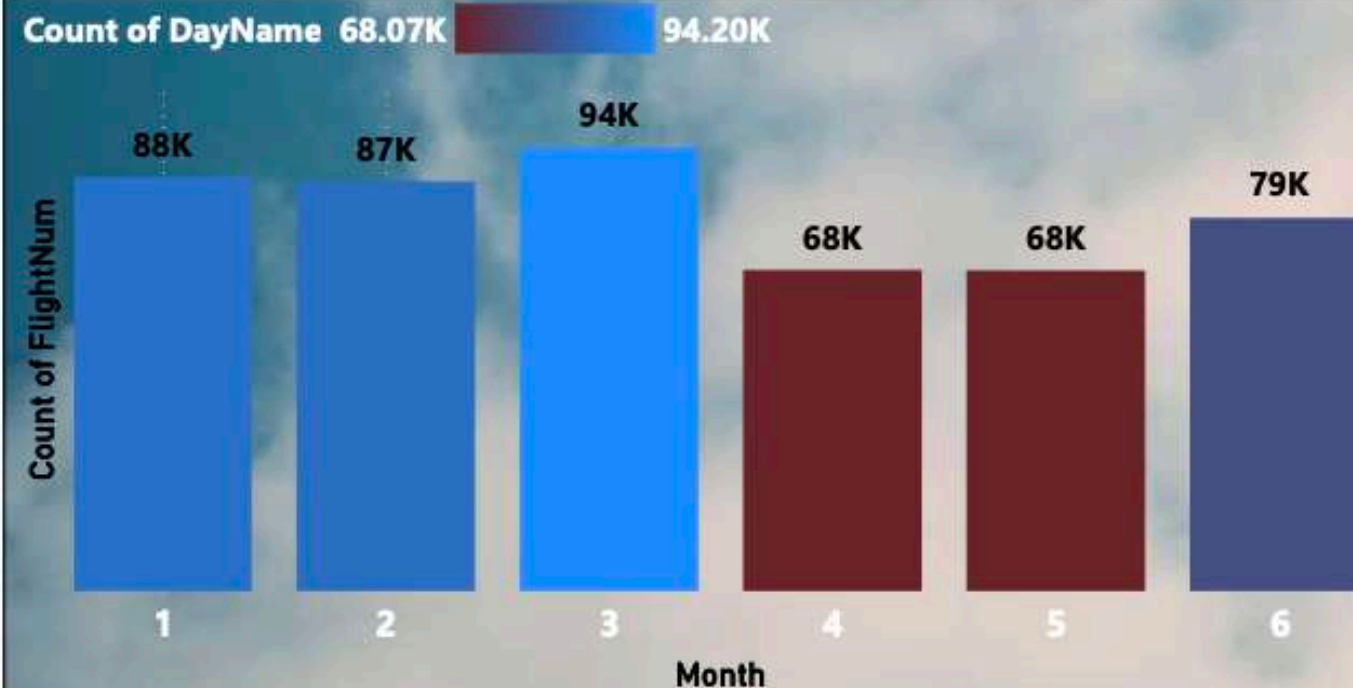
Distance by Airline



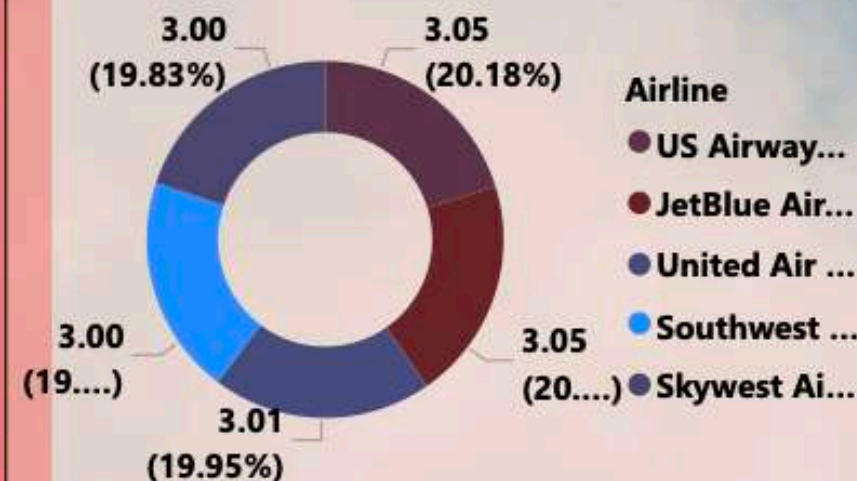
Cancelled by Airline

Alaska Airlines Inc.	0
American Airlines Inc.	0
American Eagle Airli...	0
Atlantic Southeast Ai...	0
Delta Air Lines Inc.	0
Frontier Airlines Inc.	0
Hawaiian Airlines Inc.	0
JetBlue Airways	0
Skywest Airlines Inc.	0
Southwest Airlines Co.	0
United Air Lines Inc.	0
US Airways Inc.	0

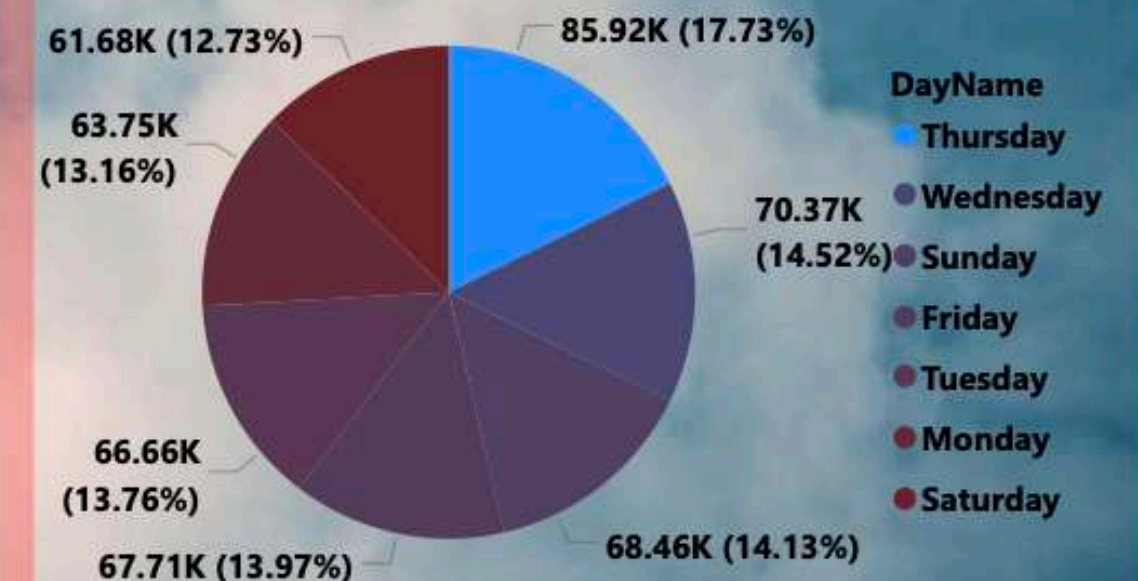
FlightNum by Month



DayOfWeekNum by Airline



FlightNum by DayName



Visualization

Delay_By_Month

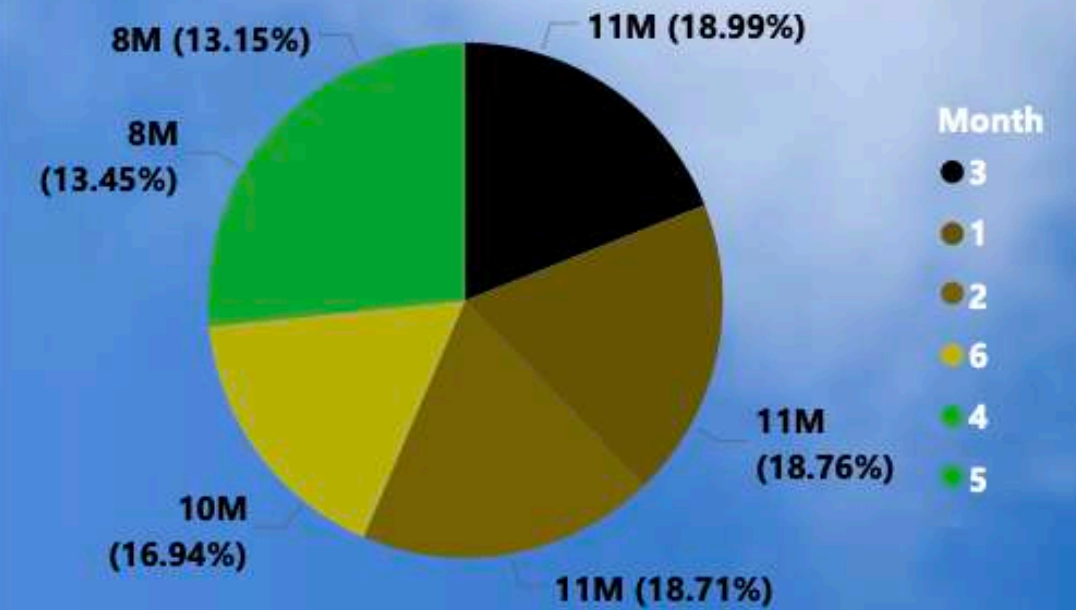
SecurityDelay by Month



WeatherDelay by Month



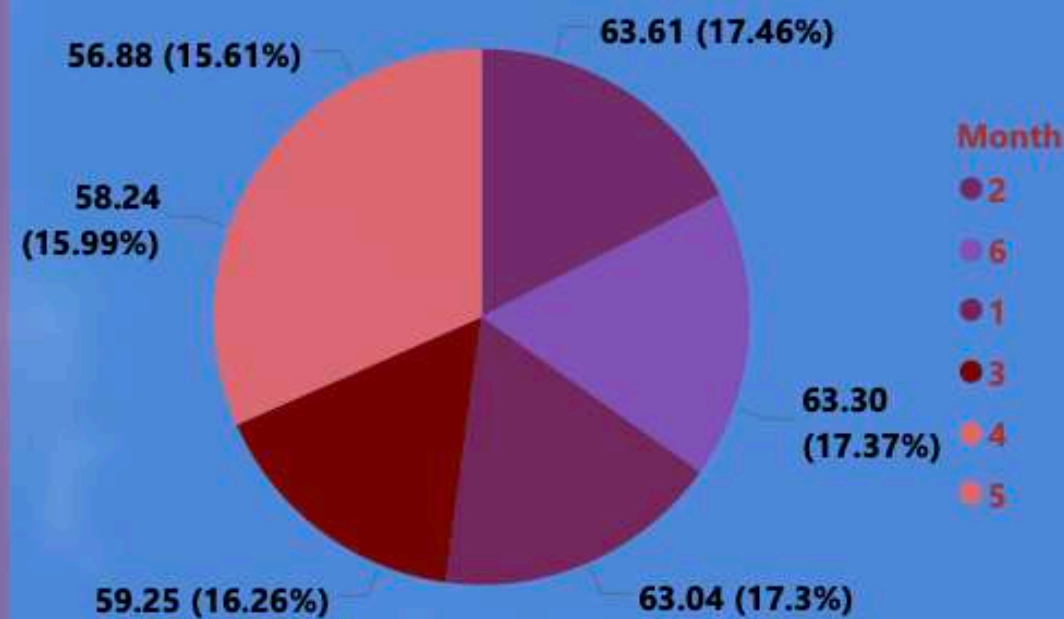
Total Delay by Month



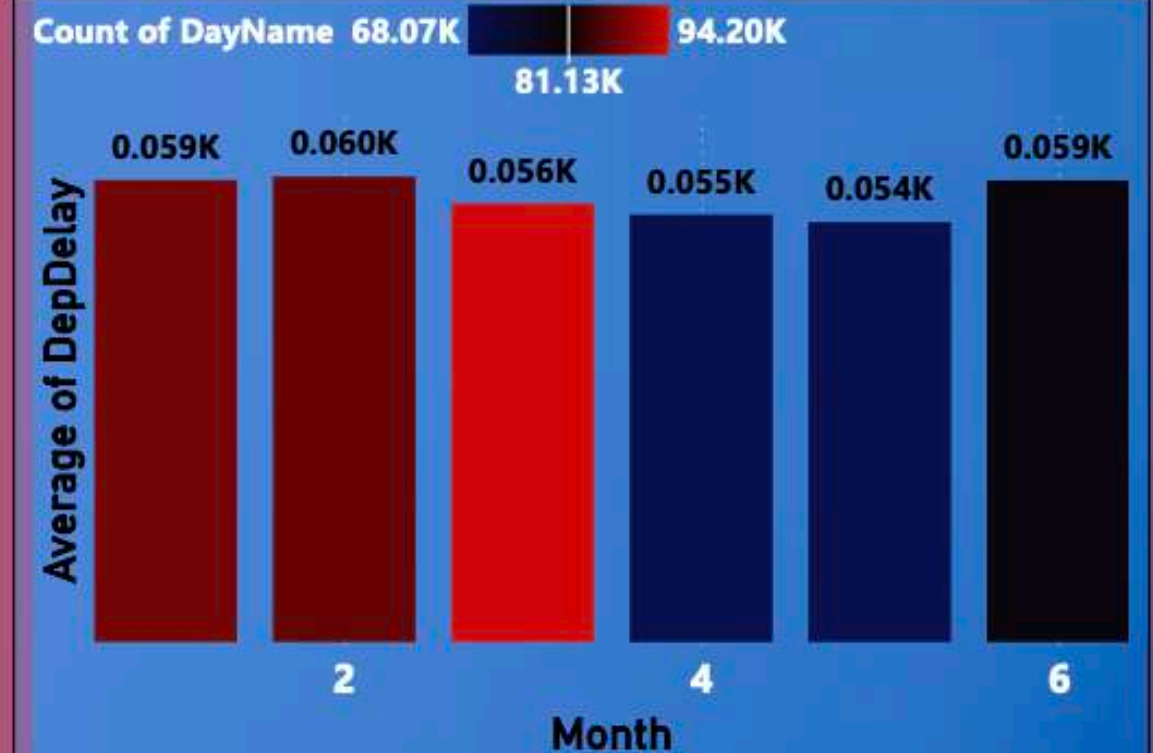
NASDelay by Month



ArrDelay by Month



DepDelay by Month



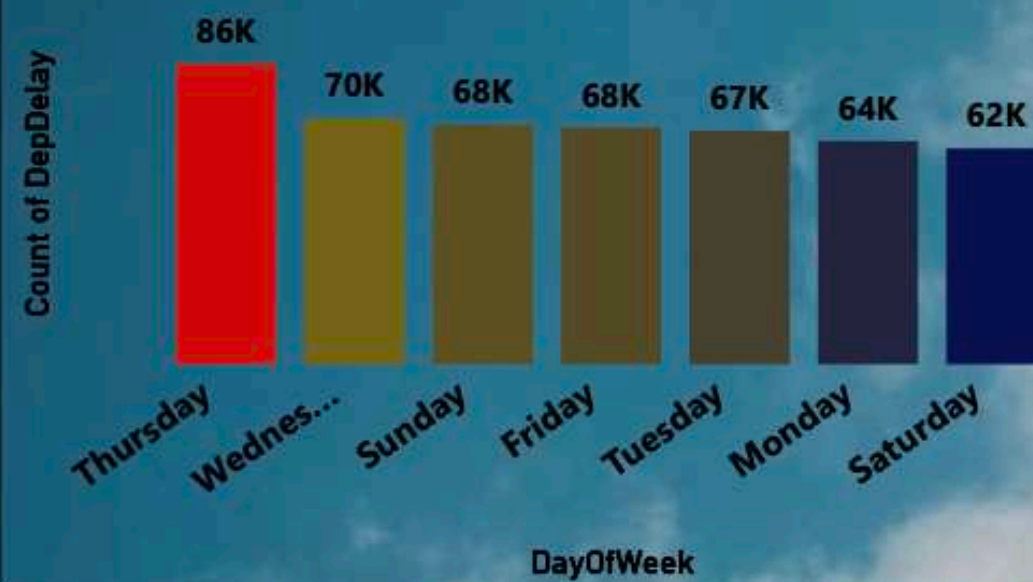
Visualization

Delay_By_DayOfWeek

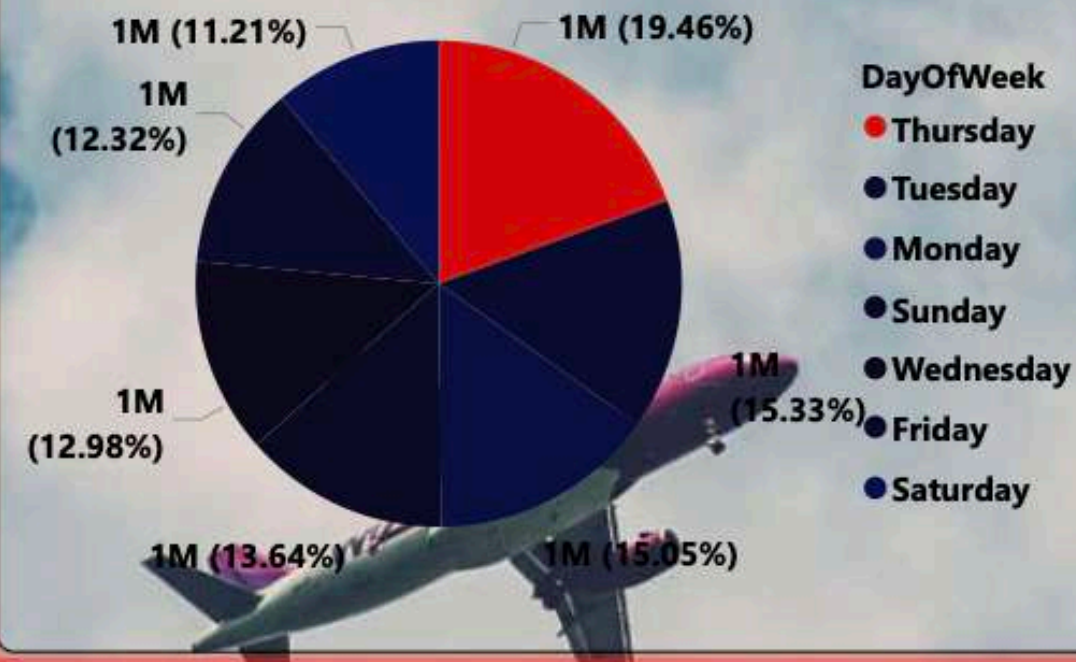
DepDelay by DayOfWeek

Count of DayName 61.68K 85.91K

73.80K



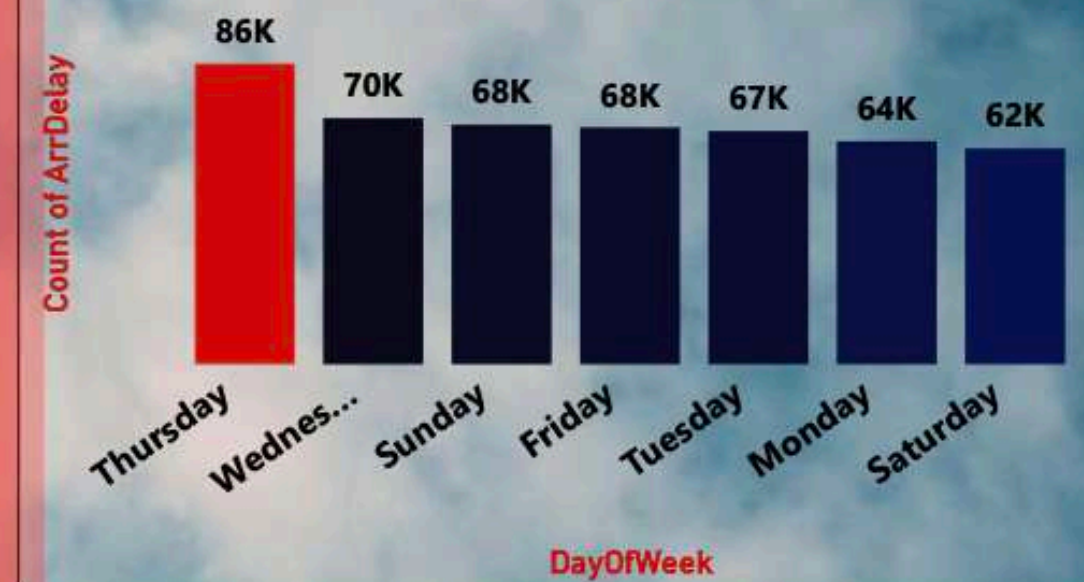
NASDelay by DayOfWeek



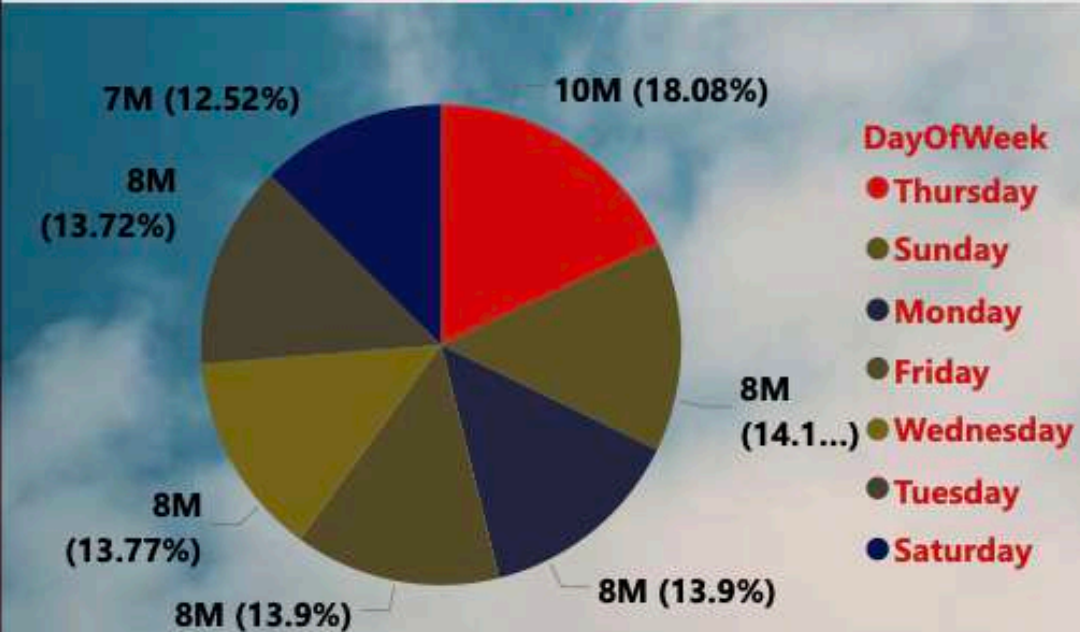
ArrDelay by DayOfWeek

Count of DayName 61.68K 85.91K

73.80K



Total Delay by DayOfWeek



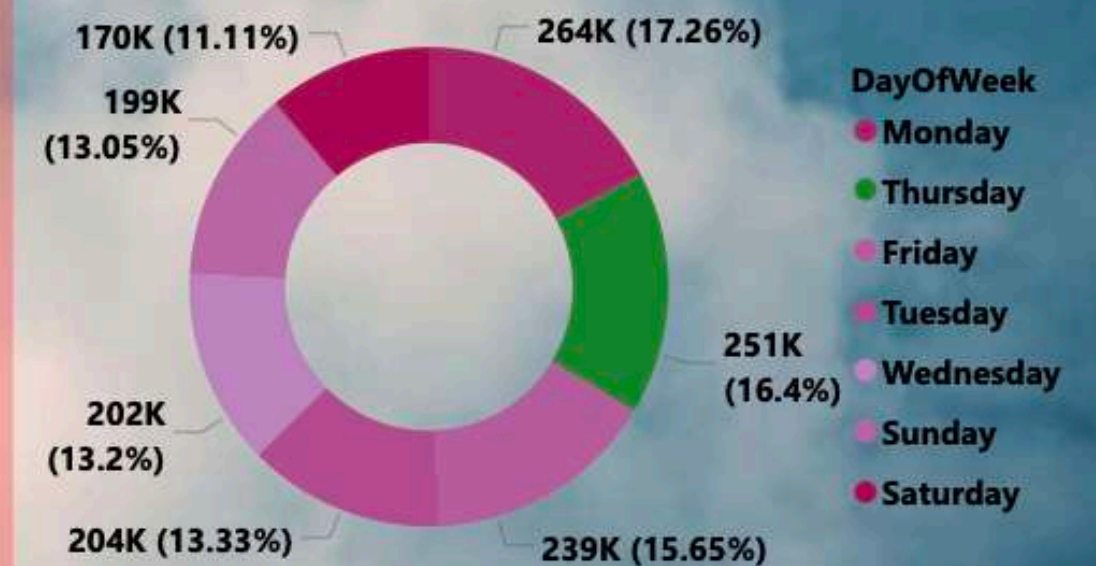
SecurityDelay by DayOfWeek

Count of DayName 61.68K 85.91K

73.80K



WeatherDelay by DayOfWeek



Challenges

During the analysis and preprocessing of the **AirFly Insights** dataset, several challenges were encountered that required careful handling to ensure accuracy, efficiency, and meaningful insights.

1. **Large Dataset Size:-** Handling over **484,000 flight records** required significant memory optimization and efficient data loading techniques to prevent processing delays.
2. **Missing and Inconsistent Data:-** Columns like *Org_Airport* and *Dest_Airport* contained **null values and irregular entries**, requiring imputation and cleaning for reliability.
3. **Text Data Standardization:-** Airline names, airport codes, and route entries varied in format, necessitating normalization for uniformity.

Challenges

4. **Multiple Delay Factors:-** Distinguishing between overlapping delay causes (*Carrier, Weather, NAS, Security, Late Aircraft*) was complex and required categorical analysis.
5. **Time-based Feature Extraction:-** Converting raw date and time fields into *Month, Day, Hour*, and *Route* features required precise parsing and error handling.
6. **Performance Optimization:-** Managing data transformations efficiently in **Pandas** to handle large-scale operations without memory overload.
7. **Complex Visualization Design:-** Creating clear, interactive **Power BI dashboards** that effectively represented multi-dimensional delay and route data.



CONCLUSION

- The project successfully transformed the raw airline delay dataset into a clean, reliable, and analysis-ready format. During preprocessing, **duplicate records (2 rows)** were completely **removed**, ensuring data uniqueness. **Text data**—including airline names, routes, and airport codes—was **cleaned and Standardized** by correcting inconsistent formats and cases. **Null values** in key columns such as *Org_Airport* (1,177) and *Dest_Airport* (1,479) were **handled effectively**, resulting in **zero missing values** in the final dataset.
- In the **delay analysis**, the **average departure delay (DepDelay)** was **8.6 minutes**, and the **average arrival delay (ArrDelay)** was **6.3 minutes**, with **taxi-out times averaging 19.15 minutes** and **taxi-in times averaging 6.78 minutes**. These results indicate operational delays were more prominent during takeoff rather than landing.
- Additionally, the **cancellation analysis** identified various causes categorized under **Carrier, Weather, NAS, and Security delays**, providing insights into major disruption factors. Overall, the cleaned and processed dataset of **484,549 records and 33 features** enabled accurate trend visualization and predictive potential for flight delay and cancellation behavior.



THANK



YOU